XRPLA

The full XRP Ledger history, compressed and verified.

Performance brief, October 2026. Measured against stock xrpld 3.4.1 and Clio 2.8 on the same machine and the same ledgers.

31×

less disk than xrpld

23 GB against 711 GB for the same 700,000 ledgers

1.8×

xrpld's WebSocket throughput

71,521 point reads per second, against 40,410

14 min

to a working xrpld

700,000 ledgers restored and verified from downloaded pieces

0

data differences from xrpld

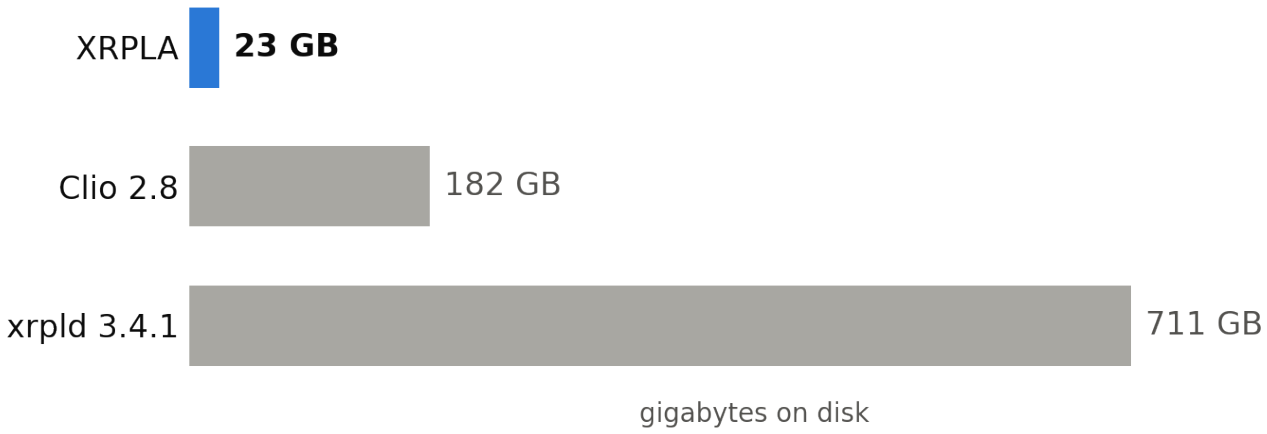
across 188,393 answers, 2013 to today

XRPLA serves the XRP Ledger's history straight from compressed, signed pieces, answers as xrpld does, and restores into xrpld and Clio. Every number in this brief comes from published raw results.

STORAGE

31× less disk than xrpld.

The same 700,000 ledgers (107,000,001 to 107,697,250), with every transaction, its metadata and the state at each ledger, as each server stores them. XRPLA keeps the history compressed and serves it without unpacking it first.



8×

less disk than Clio

Clio's Cassandra keeps a row for every object version; XRPLA keeps the changes, compressed.

17 min 06 s

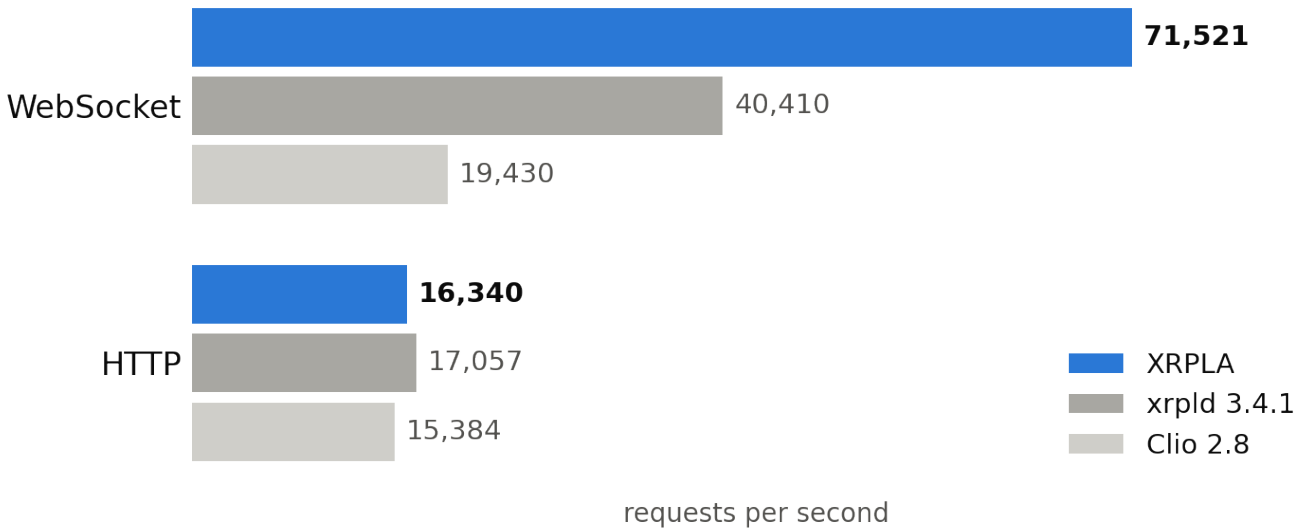
to download and index it

xrpla sync fetches the signed pieces from the public bucket and builds its indexes.

SPEED

1.8× xrpld's WebSocket throughput.

Point reads (account_info, tx, ledger, ledger_entry and server_info) across the same 700,000 ledgers, 64 requests in flight, each server on the same 24 cores of the same machine.



2.2×

xrpld's throughput per core

10,217 WebSocket reads per busy core, against 4,650.

0.86 ms

median WebSocket latency

xrpld 1.46 ms, Clio 1.25 ms.

1.7 ms

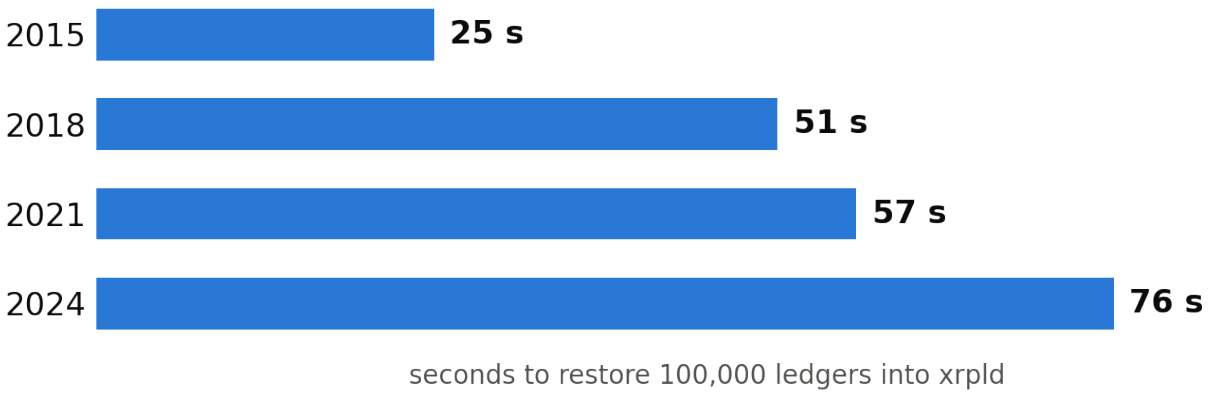
99th percentile

xrpld 2.7 ms, Clio 42.2 ms.

RESTORE

A working xrpld in 14 minutes.

xrpla restore writes xrpld's own NuDB and SQLite databases from the public pieces, checking every ledger's state root and parent chain. xrpld then starts from the restored tip, reads its state from disk rather than from peers, and joins mainnet.



seconds to restore 100,000 ledgers into xrpld

14 min 19 s

700,000 recent ledgers into xrpld

Restored and verified from pieces already downloaded, then serving mainnet from the restored tip.

952M

rows into Clio

Loaded into Cassandra in 1 h 35 min; Clio continued from the restored tip.

ACCURACY

The same answers as xrpld. Every time.

Requests derived from each sampled ledger's accounts, books, tokens and transactions, asked of XRPLA and of xrpld, compared whole, and asked again over XRPLA's WebSocket.

COMPARISON	ANSWERS	DATA DIFFERENT
Every interval of history, 2013 to today (s1, s2)	120,617	0
The 700,000 ledgers restored into xrpld	14,111	0
Older history: 100,000 ledgers of 2015	13,707	0
Older history: 100,000 ledgers of 2018	13,234	0
Older history: 100,000 ledgers of 2021	12,921	0
Older history: 100,000 ledgers of 2024	13,803	0
account_tx pages with the delegate filter	426,171	0
A Clio restored by XRPLA, against Clio's own	14,092	0

The history-wide comparison asked Ripple's public servers s1 and s2, with xrplcluster's xrpld judging any difference; the others asked xrpld 3.4.1 restored locally. On the restored range, 3 answers differed only in ping's role field, which xrpld adds for an admin connection; the Clio comparison differed only in the two nodes' version numbers.

METHODOLOGY

How we measured.

Every number in this brief is computed from the raw results published in the XRPLA repository (docs/benchmarks/2026-10-audit), with the scripts that produced them.

Machine	Hetzner: Intel Xeon Gold 5412U, 48 threads, 126 GB RAM, NVMe RAID. Each server ran on cores 16-39 and the load generator on cores 40-47.
Servers	XRPLA (a piece cache synced from the public bucket), xrpld 3.4.1 (restored by xrpla restore --to nudb and running on mainnet) and Clio 2.8.0 on Cassandra 4.1.10 (restored by xrpla restore --to clio), all holding ledgers 107,000,001 to 107,697,250.
Load	perf.py: 64 requests in flight from 4 processes, closed loop; over HTTP each request opens a new connection, over WebSocket each slot keeps one. Requests come from a corpus of recent and older ledgers of the range; each figure is the mean of the runs recorded.
Accuracy	tools/xrpld_parity.py: answers compared whole after removing only fields that describe the server. The history-wide campaign asked Ripple's public servers s1 and s2, with xrplcluster's xrpld as arbiter; the restored range and the older stretches asked xrpld 3.4.1 restored locally.
Where xrpld leads	On a mix that adds account_lines, account_objects, account_tx and book_offers at older ledgers, xrpld served 5,333 requests per second, Clio 5,603, and XRPLA 1,866 with a 32 GB decode cache (dd6d2d3).
Restore timing	The 700,000-ledger xrpld restore ran with its pieces already downloaded; the run before it, which downloaded them from the public bucket, took 15 min 46 s.
Builds	XRPLA builds of October 2026, each recorded with its raw results (the point reads: 803bc33; the full mix above: dd6d2d3); xrpld 3.4.1; Clio 2.8.0.